2025/01 / 09	*Preliminary Examination - 2025*	
X	**Mathematics (Part -II) (71)**	
Time - 2 hrs.	**pages - 08**	**Marks : 40**

Note : -
- **i) All questions are compulsory.**
- **ii) Use of calculator is not allowed.**
- **iii) The numbers to the right of the questions indicate full marks.**
- **iv) In case of MCQs [(Q.No.1(A)] only the first attempt will be evaluated and will be given credit.**
- **v) For every MCQ, the correct alternative (A),(B),(C) or (D) with subquestion number is to be written as an answer. for Eg- i) (A), ii) (B) , iii) (C)**
- **vi) Draw the proper figures for answers wherever necessary.**
- **vii) The marks of construction should be clear and distinct.Do not erase them.**
- **viii) Diagram is essential for writing the proof of the theorem.**

Q.1 A) Four alternative answers are given for every subquestion select the correct alternative and write the alphabet of that answer. **04**

i) If l,m,n are sides of a triangle and $l^2 + m^2 = n^2$,name the type of triangle.

A) Obtuse angled triangle B) Acute angled triangle

C) Right angled triangle D) Equilateral triangle

ii) Two circles of radii 5.5cm and 3.3 cm respectively touch each other. What is the distance between their centres?

A) 4.4cm B) 8.8cm

C) 2.2 cm D) 8.8 or 2.2 cm

iii) Out of the following point .....lies to the right of the origin on x -axis.

A) (-2 , 0) B) (0 , 2)

C) (2, 3) D) (2,0)

iv) Find the side of a cube of volume $1m^3$

A) 1cm B) 10 cm

C) 100 cm D) 1000 cm

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Q.1 B) Solve the following sub-questions. 04

1) $\Delta LMN \sim \Delta PQR$, $A(\Delta LMN) = 9$, $A(\Delta PQR) = 36$

then fin the value of ratio $\frac{LM}{PQ}$

2) In ΔADC, $\angle D = 90^0$, $\angle C = 45^0$, $AD = 8$ then find AC.

3) If radius of a circle is 7cm,then find the length of longest chord of a circle.

4) Find the distance between the points A(-5,3) , B (5,3)

Q.2 A) Complete the following activities.(any two) 04

i) In figure along side two circles intersect each other in points X and Y. Tangents drawn from a point M on line XY touch the circles at P and Q. Prove that,seg PM $\cong$ seg QM

Fill in the blanks and write proof.

Line MX is a common [. ....] of the two circles.

$\therefore PM^2 = MY \times MX$ .....,.....(I)

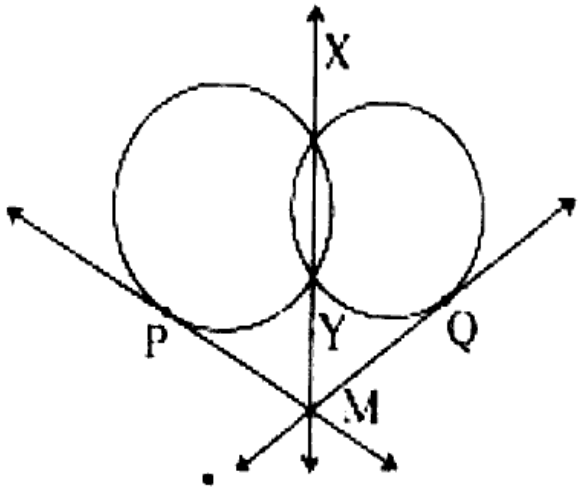
Similarly QM^2 = [..... .....] $\times MX$,

tangent secant segement theorem .....(II)

$\therefore$ From (I) and (II) [..... ..] $= QM^2$

$\therefore PM = QM$

seg PM $\cong$ [ ]



ii) Show that $\frac{\tan\theta}{\sec\theta+1} = \frac{\sec\theta-1}{\tan\theta}$

Proof :

$$\text{LHS} = \frac{\tan\theta}{\sec\theta+1}$$

$$= \frac{\tan\theta \times \tan\theta}{(\sec\theta+1) \times \square}$$

$$= \frac{\square}{(\sec\theta+1) \times \tan\theta}$$

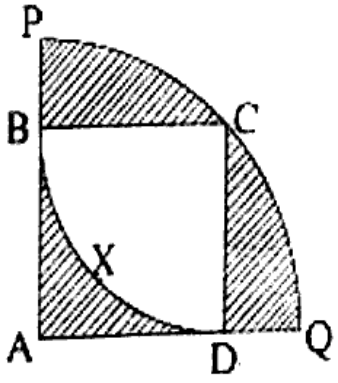
$$= \frac{\square(\quad)}{(\sec\theta+1) \times \tan\theta} \quad \ldots\ (1+\tan^2\theta = \sec^2\theta)$$

$$= \frac{(\sec\theta-1)\ (\square)}{\tan\theta+1 \times \tan\theta} \quad \ldots\ a^2-b^2 = (a+b)(a-b)$$

$$= \frac{(\sec\theta-1)}{\tan\theta} = \text{RHS}$$

$\therefore$ LHS = RHS Hence proved

ii) In the figure,square ABCD is inscribed in the sector A- PCQ. The radius of sector C-BXD is 20 cm.Complete the following activity to find the area of shaded region inside the square.



Solution : Side of square ABCD = radius of sector C- BXD = 20cm

Area of square = (side) ^{2} = (20) ^{2} = $\square$

Area of shaded region inside the square.

= Area of square ABCD - Area of sector C - BXD

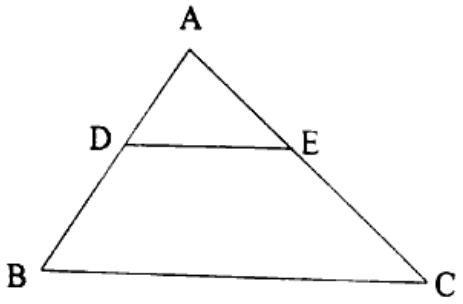
$= (CD)^2 - \frac{\theta}{360} \times$ [ ]

$= 400 - \frac{90}{360} \times \frac{3.14}{1} \times \frac{400}{1}$

$= 400 -$ [ ] $=$ [ ]

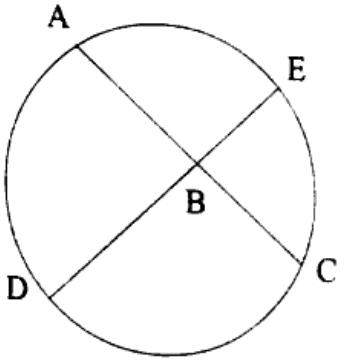
B) Solve the following sub - questions (any four) 08

1) In Δ ABC, DE // BC
If DB = 5.4 cm, AD = 1.8cm
EC = 7.2 cm then find AE.



2) Find the length of the hypotenuse of a right angled triangle if remaining sides are 9cm and 12 cm .

3) In adjacent figure chords AC and DE inersect at B.If
$\angle ABE = 108^0$, m (arc AE) = 95^0,
Find m(arc DC)



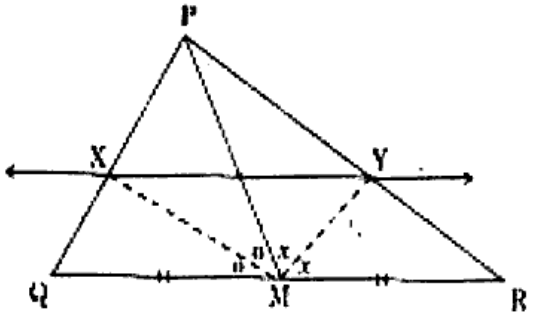
4) Show that points A(-1,-1), B(0,1), C(1,3) are collinear.

5) A person is standing at a distance of 80 m from a building looking at its top. The angle of elevation is of 45^0. Find the height of the building.

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Q.3 A) Complete the following activities.(any one) 03

i)



In ΔPQR, seg PM is a median. Angle bisectors of $\angle PMQ$ and $\angle PMR$ intersect side PQ and side PR in points X and Y respectively.

Prove that XY $\parallel$ QR

Complete the proof by filling in the boxes.

Solution :

In ΔPMQ

Ray MX is the bisector of $\angle PMQ$

$\therefore \frac{MP}{MQ} = \frac{\square}{\square}$ .....(I) [Theorem of angle bisector]

Similarly, in ΔPMR , Ray MY is bisector of $\angle PMR$

$\therefore \frac{MP}{MR} = \frac{\square}{\square}$ .....(II) [Theorem of angle bisector]

But $\frac{MP}{MQ} = \frac{MP}{MR}$ .....(III) [As M is the midpoint of QR]

Hence MQ = MR

$\therefore \frac{PX}{\square} = \frac{\square}{YR}$ ..... [From (I) , (II) , (III)]

$\therefore$ XY $\parallel$ QR .....[Converse of basic proportionality theorem.]

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ii) If point T divides the segment AB with A(-7,4) and B(-6,-5) in the ratio 7:2,find the co-ordinates of T.

Let the co-ordinates of T be (x,y)

$\therefore$ by the section formula,

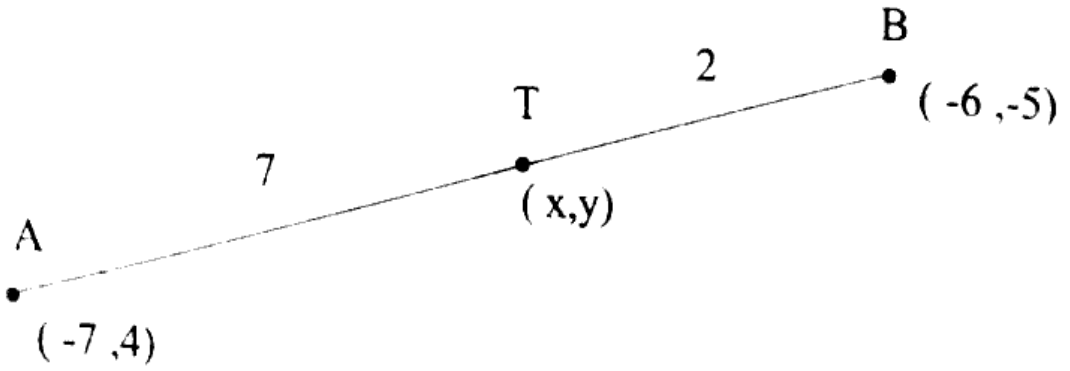
$$x = \frac{mx_2 + nx_1}{m+n} = \frac{7 \times (-6) + \boxed{\quad \times \quad}}{7+2}$$

$$= \frac{-42 - 14}{9} = \frac{\boxed{\quad}}{9}$$

$$y = \frac{my_2 + ny_1}{m+n} = \frac{\boxed{\quad \times (\quad)} + 2(4)}{7 + \boxed{\quad}}$$

$$= \frac{\boxed{\quad}}{9} = \frac{-27}{9} =$$

$\therefore$ Co -ordinates of point T are $\boxed{(\quad , \quad)}$



Q.3 B) Solve the following sub-questions (any two) **06**

1) In Δ PQR, PQ = 10, PR = 7, QR = 9 then find the length of the median drawn from R to side PQ.

2) prove that, " An exterior angle of a cyclic quadrilateral is congruent to the angle opposite to its adjacent interior angle."

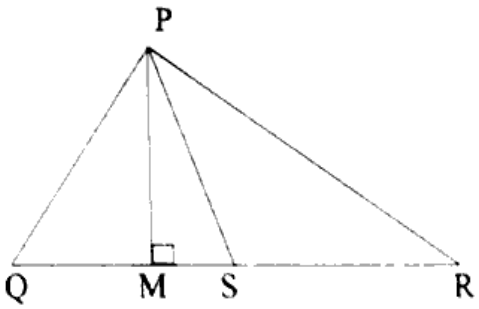
3) Draw a circle with centre P.Draw an arc XY of measure 105^0. Draw tangents to the circle at point X and Y.

4) Radius of a circle is 10cm.Area of the sector is $100cm^2$.Find the area of its corresponding major sector. ($\Pi = 3.14$)

Q.4 Solve the following sub-questions (any two) **08**

1) In Δ PQR, PS is the bisector of $\angle$ QPR.Seg PM $\perp$ side QR
Then prove that :
A (Δ PQS): A(Δ PRS) = PQ: PR.

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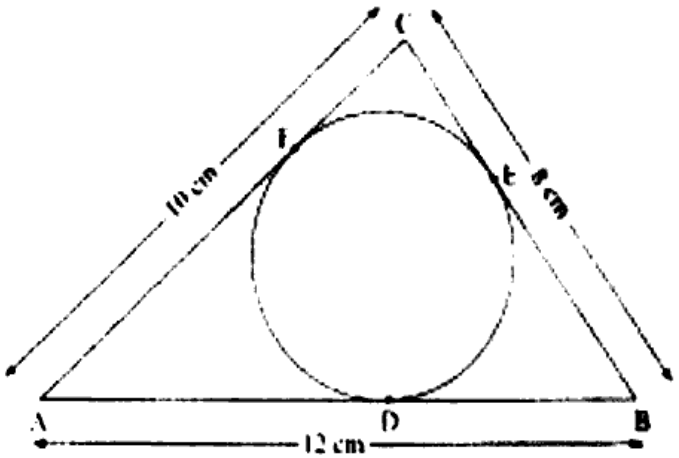
2) ΔPSE $\sim$ Δ TSV.In ΔPSE, PS = 4.5 cm, SE = 5.2cm,PE = 5.4cm and PS : TS = 5:3 . Construct ΔPSE and Δ TSV.

3) An ice -cream pot has right circular cylindrical shape.The radius of the base is 16 cm and height is 14 cm.This pot is completely filled with ice -cream.The entire ice-cream is given to the students in the form of rihgt circular ice -cream cones,having diameter 4cm and height is 7cm. If each student is given one cone, how many students can be served?

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Q.5 Solve the following sub-questions (any one) **03**

1) In the figure, a circle is inscribed in ΔABC such that it touches the sides AB, BC and CA at points D,E and F respectively.If AB =12cm BC = 8 cm and AC = 10 cm then find the length of AD.



2) Find the value of $9\tan^2 x - \frac{9}{\cos^2 x}$